CVT System Model Derivation

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Contents

1 Overview

1.1 Purpose of This Document

This document presents a complete dynamic derivation of a continuously variable transmission (CVT) system model. The goal is to develop a physically consistent set of governing equations that describe:

- Rotational dynamics of the primary and secondary assemblies,
- Axial shift dynamics of the primary sheaves,
- Belt traction and torque transfer,
- Slip behavior under torque saturation,
- Coupled vehicle longitudinal response.

The resulting formulation provides a structured, state-based model suitable for numerical simulation and performance analysis.

1.2 Scope

This document focuses on the mechanical modeling of the CVT system. It includes force balances, torque transfer mechanisms, and dynamic coupling between rotational and axial motion.

The following are outside the scope of this document:

- Thermal modeling,
- Long-term wear or material degradation,
- Detailed tribological contact mechanics,
- Multi-body vehicle dynamics beyond one-dimensional motion.

All assumptions limiting model fidelity are formally defined in Section 5.

1.3 Expected Background Knowledge

The reader is expected to be familiar with:

- Newton's laws of motion,
- Basic rigid-body rotational dynamics,
- Linear spring models,
- Gear ratios and torque multiplication,
- Basic algebra and differential equations.

Prior familiarity with state-space modeling or numerical integration is not assumed. Relevant concepts are introduced where required.

1.4 Conceptual Overview of a CVT

A continuously variable transmission (CVT) transmits torque between an engine and drivetrain using a belt or chain running between two variable-diameter pulleys.

Each pulley consists of two opposing conical faces (sheaves) whose axial separation determines the effective belt radius. As the primary sheaves shift axially, the effective radius changes, which in turn modifies the transmission ratio.

Unlike discrete gearboxes, a CVT provides a continuously varying ratio. Torque transfer is governed by belt traction, normal clamping forces, and frictional interaction between the belt and sheave faces.

When transmitted torque exceeds traction capacity, slip occurs. Accurately modeling both stick and slip regimes is essential for predicting performance.

1.5 Modeling Approach

The system is modeled using a minimal set of dynamic state variables, defined in Section 7. All other quantities are derived algebraically from these states.

The derivation proceeds as follows:

- Governing physical laws are defined (Section 6),
- Kinematic relationships between pulley geometry and ratio are derived,
- Axial and radial force models are constructed,
- Belt traction limits are formulated,
- Rotational and axial equations of motion are assembled,
- Slip behavior is incorporated.

The final result is a coupled nonlinear dynamic system suitable for time-domain simulation.

2 Notation and Conventions

2.1 Coordinate and Sign Conventions

A consistent set of coordinate directions and sign conventions is adopted throughout this document. All equations follow these conventions.

2.1.1 Vehicle Longitudinal Direction

The vehicle is modeled as a one-dimensional system moving along a single longitudinal axis.

- Positive direction ($+x$) is defined as forward vehicle motion.
- Vehicle velocity $v > 0$ corresponds to forward motion.
- Resistive forces (rolling resistance, aerodynamic drag, grade load) act in the negative ($-x$) direction.

Incline Angle If terrain grade is considered, the incline angle α is defined as:

- $\alpha > 0$ for uphill motion.
- $\alpha = 0$ for level ground.

The gravitational component acting along the longitudinal axis is:

$$F_{\text{grade}} = mg \sin \alpha$$

This modifies the longitudinal load only and does not introduce additional spatial dimensions into the model.

2.1.2 Rotational Sign Convention

All angular velocities follow the same convention:

- Positive angular velocity corresponds to forward vehicle motion.
- Primary and secondary pulleys are defined such that $\omega_p > 0$ and $\omega_s > 0$ produce forward belt motion.
- Applied engine torque is positive when it increases ω_p.

Reaction torques opposing motion are therefore negative.

2.1.3 CVT Axial Direction

The axial direction of the primary pulley is defined along its shaft axis.

- Positive axial displacement $s > 0$ corresponds to increasing primary effective radius.
- Increasing s therefore corresponds to an upshift (decreasing overall CVT ratio).

The axial velocity $\dot{s}$ follows the same sign convention.

2.1.4 Radial Direction

Radial direction is defined perpendicular to the pulley axis, measured outward from the center of rotation.

- Radial forces $F_r > 0$ act outward from the pulley center.
- Belt normal forces are treated as positive when compressive.

2.1.5 Slip Velocity Sign Convention

Slip velocity between belt and pulley is defined as:

$$v_{\text{slip}} = r_p(s)\omega_p - r_s(s)\omega_s$$

- $v_{\text{slip}} > 0$ indicates the primary surface speed exceeds the secondary surface speed.
- $v_{\text{slip}} = 0$ corresponds to no-slip contact.

2.2 Symbol Summary

Symbol	Meaning
v	vehicle speed
ω_e	engine angular speed
s	primary axial shift position
$\dot{s}$	primary axial shift velocity
R	CVT ratio (define direction explicitly)

Table 1: Core symbols used throughout the derivation.

3 Units and Dimensions

All quantities in this document are expressed in SI units unless otherwise stated.

3.1 Base Units

Quantity	Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Plane angle	radian	rad

Table 2: SI base units used throughout the model.

3.2 Common Derived Units

Quantity	Unit	Symbol
Velocity	meter per second	m/s
Acceleration	meter per second squared	m/s^2
Angular velocity	radian per second	rad/s
Angular acceleration	radian per second squared	rad/s^2
Force	newton	N
Torque	newton meter	N m
Moment of inertia	kilogram meter squared	$kg\,m^2$
Power	watt	W

Table 3: Common derived SI units used in the CVT model.

4 Constants and Parameters

All physical constants, geometric parameters, and calibrated values used in the model are listed below.

Symbol	Name	Value	Unit	Description
m	Vehicle mass	–	kg	Total vehicle mass
r_w	Wheel radius	–	m	Effective rolling radius
G	Final drive ratio	–	–	Secondary-to-wheel speed ratio
I_p	Primary inertia	–	$kg\,m^2$	Lumped primary-side inertia
I_s	Secondary inertia	–	$kg\,m^2$	Lumped secondary-side inertia
k_p	Primary spring constant	–	N/m	Primary compression spring
k_s	Secondary spring constant	–	N/m	Secondary compression spring
μ	Friction coefficient	–	–	Belt-pulley friction coefficient
β	Wrap angle	–	rad	Belt wrap angle

Table 4: Model constants and parameters.

5 Modeling Assumptions

The following assumptions define the modeling scope and physical idealizations used in the CVT dynamic formulation. These assumptions establish the limits within which the derived equations remain valid.

5.1 Material and Structural Idealizations

Assumption 1 (Constant Material Properties)

Material properties of structural components are treated as constant. No temperature-dependent material variation is modeled. Thermal effects are not solved as part of this system.

Assumption 2 (Rigid Structural Components)

All structural components (pulleys, shafts, housing) are modeled as rigid bodies. Elastic deformation under operational loading is neglected. The belt is assumed axially rigid (no longitudinal stretch), while still conforming geometrically around pulleys.

Assumption 3 (Negligible Wear and Surface Evolution)

Component wear, belt degradation, and pulley surface evolution are neglected. The model represents short-term operational behavior.

5.2 Contact and Friction Modeling

Assumption 4 (Idealized Belt–Pulley Friction Law)

Belt traction is modeled using a simplified friction law (e.g., Coulomb friction with a capstan formulation and an explicit stick–slip transition rule). Microscopic contact mechanics, rubber hysteresis, lubrication films, and detailed Stribeck effects are not explicitly modeled.

Assumption 5 (Constant Effective Friction Coefficient)

The belt–pulley coefficient of friction is treated as constant. It does not evolve with temperature, wear, or surface conditioning.

Assumption 6 (Uniform Contact Distribution)

Contact pressure and traction over the pulley wrap angle are treated as effectively uniform or averaged. Localized edge loading and non-uniform stress distributions are neglected.

5.3 Geometric and Kinematic Idealizations

Assumption 7 (Perfect Axis Alignment)

Primary and secondary pulley axes are assumed perfectly aligned. Misalignment, shaft eccentricity, and geometric runout are neglected.

Assumption 8 (Constant Belt Length)

The belt length is treated as constant. Permanent creep and progressive elongation are not modeled. Effective pulley radii are determined solely from geometry and belt-length constraints.

5.4 Dynamic and Inertial Modeling

Assumption 9 (Lumped Inertia Representation)

Rotational inertias of the engine, pulleys, drivetrain, and wheels are represented as lumped equivalent inertias. Distributed shaft flex and torsional compliance are neglected.

Assumption 10 (Negligible Vibrational Effects)

High-frequency vibrational effects, structural resonances, and drivetrain oscillations are neglected. The model captures dominant rigid-body dynamics only.

Assumption 11 (Negligible Parasitic Friction)

Frictional losses in non-critical components (e.g., bearings, seals, auxiliary shafts) are neglected. Only friction directly relevant to belt traction and CVT torque transfer is modeled.

5.5 Environmental and Operational Scope

Assumption 12 (Negligible Gravitational Influence on CVT Internals)

Gravitational forces acting on CVT internal components are negligible relative to centrifugal and contact forces at operating speeds.

Assumption 13 (Full-Throttle Engine Model)

The engine is modeled using a prescribed full-throttle torque curve. Load feedback modifies engine speed dynamically, but part-throttle dynamics and throttle transients are not explicitly modeled.

6 Governing Principles

Theory 1: Newton's Second Law (Translation)

Equation

$$\sum F = ma$$

Description

- $\sum F$ is the net force (N)
- m is mass (kg)
- a is acceleration (m/s^2)

Source

Classical mechanics (textbook / lecture notes).

Derivation

From the definition of acceleration and empirical laws of motion; used here as a governing relationship.

Theory 2: Newton's Second Law (Rotation)

Equation

$$\sum \tau = I\alpha$$

Description

- $\sum \tau$ is net torque (N m)
- I is moment of inertia ($\mathrm{kg\,m^2}$)
- α is angular acceleration ($\mathrm{rad/s^2}$)

Source

Classical rigid body dynamics (textbook / lecture notes).

Derivation

Standard rigid-body rotational dynamics relationship.

Theory 3: Centrifugal Force

Equation

$$F_c = m\omega^2 r$$

Description

- F_c is centrifugal force (N)
- m is mass (kg)
- ω is angular velocity (rad/s)
- r is radius from the axis of rotation (m)

Source

Bogna Szyk and Steven Wooding (2024).

Derivation

Derived from circular motion kinematics where $a_c = \omega^2 r$, then applying $F = ma$.

Theory 4: Hooke's Law (Compressional)

Equation

$$F = k\Delta x$$

Description

- F is force (N)
- k is spring constant (N/m)
- Δx is displacement (m)

Source

Encyclopaedia Britannica (2024).

Derivation

Linear spring constitutive relationship in the elastic regime.

Theory 5: Hooke's Law (Torsional)

Equation

$$F = \frac{k\Delta\theta}{r}$$

Description

- F is force (N)
- k is torsional spring constant (N m/rad)
- $\Delta\theta$ is angular deflection (rad)
- r is torque arm radius (m)

Source

Dr. Templeman (2024).

Derivation

Starting from $\tau = k\Delta\theta$ and $\tau = Fr$, then $F = \frac{k\Delta\theta}{r}$.

Theory 6: Engine Torque from Power

Equation

$$\tau = \frac{P}{\omega}$$

Description

- τ is torque (N m)
- P is power (W)
- ω is angular velocity (rad/s)

Source

2024 Power Test, LLC (2024).

Derivation

From the rotational power relationship $P = \tau\omega$.

Theory 7: Gearing (Torque Scaling)

Equation

$$\tau_{\text{out}} = \tau_{\text{in}}\, R_{\text{gear}}$$

Description

- τ_{out} is output torque (N m)
- τ_{in} is input torque (N m)
- R_{gear} is gear reduction ratio (unitless)

Source

Brain (2023).

Derivation

Ideal gear relationship assuming no losses; power balance implies inverse scaling of speed.

Theory 8: Static Friction

Equation

$$F_f = \mu_s F_n$$

Description

- F_f is friction force (N)
- μ_s is coefficient of static friction (unitless)
- F_n is normal force (N)

Source

Wikipedia contributors (2024).

Derivation

Coulomb friction model for impending slip.

Theory 9: Capstan Equation

Equation

$$T_1 = T_2 e^{\mu\theta}$$

Description

- T_1 is tight-side belt tension (N)
- T_2 is slack-side belt tension (N)
- μ is belt–pulley friction coefficient (unitless)
- θ is wrap angle (rad)

Source

Hackaday (2021).

Derivation

Differential force balance on an infinitesimal belt element on a pulley yields $\frac{\mathrm{d}T}{T} = \mu \, \mathrm{d}\theta$, then integrate over wrap.

Theory 10: Density

Equation

$$\rho = \frac{m}{V}$$

Description

- ρ is density (kg/m^3)
- m is mass (kg)
- V is volume (m^3)

Source

The Editors of Encyclopaedia Britannica (2024).

Derivation

Definition of density.

Theory 11: Air Resistance (Quadratic Drag)

Equation

$$F_d = \frac{1}{2}\rho v^2 C_d A$$

Description

- F_d is drag force (N)
- ρ is fluid density (kg/m^3)
- v is object speed relative to fluid (m/s)
- C_d is drag coefficient (unitless)
- A is reference area (m^2)

Source

SoftSchools (2020).

Derivation

Standard empirical/derived quadratic drag model; applicable at moderate to high Reynolds number.

7 System State Definition

The dynamic behavior of the CVT system is described using a set of *state variables*. A state variable is a quantity whose future value depends on its current value and its time derivative. In numerical simulation, state variables are the quantities that are directly integrated forward in time at each timestep.

All other quantities in the model (e.g., torque transfer, CVT ratio, belt forces) are computed algebraically from these states.

7.1 State Vector

State Vector

$$\boldsymbol{x}(t) = \begin{bmatrix} \omega_p \\ \omega_s \\ s \\ \dot{s} \end{bmatrix}$$

where:

- ω_p is the angular speed of the primary pulley (rad/s).
- ω_s is the angular speed of the secondary pulley (rad/s).
- s is the axial shift position of the primary sheave (m).
- $\dot{s}$ is the axial shift velocity (m/s).

7.2 Why These Are the State Variables

These four variables represent the independent dynamic degrees of freedom of the CVT system:

- The primary pulley rotation (ω_p) evolves according to applied engine torque and belt reaction torque.
- The secondary pulley rotation (ω_s) evolves according to belt torque and external load torque.
- The axial shift position (s) evolves according to the balance of axial forces between the primary and secondary assemblies.
- The axial velocity ($\dot{s}$) is required because axial motion follows a second-order differential equation.

This will be made clear in the following derivations in later sections (TODO: MAKE BETTER). Removing any of these variables would eliminate an independent dynamic degree of freedom from the model.

7.3 Numerical Integration Perspective

At each simulation timestep Δt, the following updates occur:

- ω_p is updated using its angular acceleration.
- ω_s is updated using its angular acceleration.
- s is updated using $\dot{s}$.
- $\dot{s}$ is updated using axial acceleration.

Thus, the state vector evolves according to differential equations derived in later sections.

All other model quantities — including CVT ratio, belt slip velocity, radial forces, and torque capacity — are computed directly from the current values of ω_p, ω_s, and s.

8 Pulley Geometry

This section establishes the geometric relationships that govern how axial shift modifies effective pulley radii and therefore transmission ratio. The derivations presented here are purely kinematic and do not involve force balances.

The objective is to derive explicit expressions for:

- Primary outer radius as a function of shift,
- Secondary outer radius as a function of shift,
- Center-to-center distance,
- CVT ratio and its rate of change.

8.1 Conceptual Overview of Variable Geometry

Each pulley in the CVT consists of two opposing sheaves. One sheave is fixed axially, while the other is free to translate along the shaft.

The effective belt radius is determined by the axial separation between these sheaves. As the moveable sheave approaches the fixed sheave, the belt is forced outward along the conical faces, increasing the effective pulley radius.

Conversely, as the sheaves separate, the belt rides deeper between the sheaves, decreasing the effective radius.

Because the belt length is constant, any increase in the primary pulley radius must be accompanied by a corresponding decrease in the secondary pulley radius. This geometric coupling is fundamental to the operation of the CVT and is derived explicitly in the following sections.

TODO: ADD ILLUSTRATION. Cvt cross section of single pulley in 0 shift and full shift

8.2 Definition of Axial Shift Variable

Let s denote the axial displacement of the moveable primary sheave, defined according to the sign convention in Section 2.1.

- At $s = 0$, the primary sheaves are at their maximum axial separation.
- At $s = 0$, the primary pulley is at its minimum effective radius.

For $s > 0$, the moveable sheave translates toward the fixed sheave, reducing the axial gap between the sheaves.

Deadzone and Travel Limits The shift coordinate is bounded as

$$0 \le s \le s_{\text{max}}.$$

A deadzone region exists such that

$$0 \le s < s_{\text{dz}}$$

does not alter the belt contact radius on the sheaves.

For

$$s_{\mathrm{dz}} \le s \le s_{\max},$$

axial motion produces a proportional change in primary radius.

Here:

- s_{dz} denotes the deadzone displacement,
- $s_{\max}$ denotes the maximum allowable sheave travel.

8.3 Primary Outer Radius as a Function of Shift

Given sheave half-angle β, axial closure beyond the deadzone produces a radial increase governed by simple trigonometry.

For an incremental axial displacement Δs beyond s_{dz}, the corresponding radial increase is

$$\Delta r = \frac{\Delta s}{2\tan\beta}.$$

The factor of 2 arises because axial closure is shared symmetrically between the two opposing sheave faces. The geometric relation is illustrated in Figure 1.

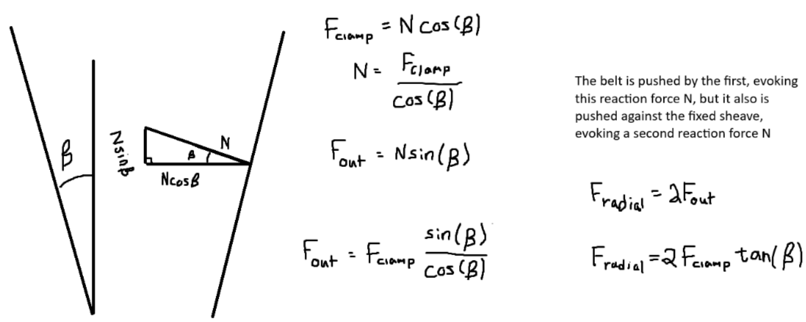

Figure 1: Axial–radial relationship for opposing conical sheaves with half-angle β.

Piecewise Radius Model Let $r_{p,\min}$ denote the minimum primary outer radius. Then the primary outer radius is

$$r_{p,\mathrm{outer}}(s) = \begin{cases} r_{p,\min}, & 0 \le s < s_{\mathrm{dz}}, \\ r_{p,\min} + \dfrac{s - s_{\mathrm{dz}}}{2\tan\beta}, & s_{\mathrm{dz}} \le s \le s_{\max}. \end{cases} \tag{8.1}$$

Remark on Belt Height and Effective Torque Radius

The belt possesses a finite radial thickness h_b. In this section, all geometric relations are formulated using the *outer surface* of the belt as the reference radius. That is, r_{outer} denotes the radial position of the belt's outer face.
This convention simplifies the geometric derivations of pulley radius and belt length. However, torque transmission does not occur at the outer surface. The effective moment arm corresponds approximately to the mid-height of the belt.
Accordingly, the effective torque radius used in force and torque calculations will later be defined as

$$r_{\text{eff}} = r_{\text{outer}} - \frac{h_b}{2}. \tag{8.2}$$

This correction is applied explicitly in the torque transfer formulation (see Section 8.7).

8.4 Belt Length Constraint

This section derives the geometric constraint relating belt length to the primary radius r_p, secondary radius r_s, and center-to-center distance C. The derivation follows directly from the belt path geometry shown in Figure 2.

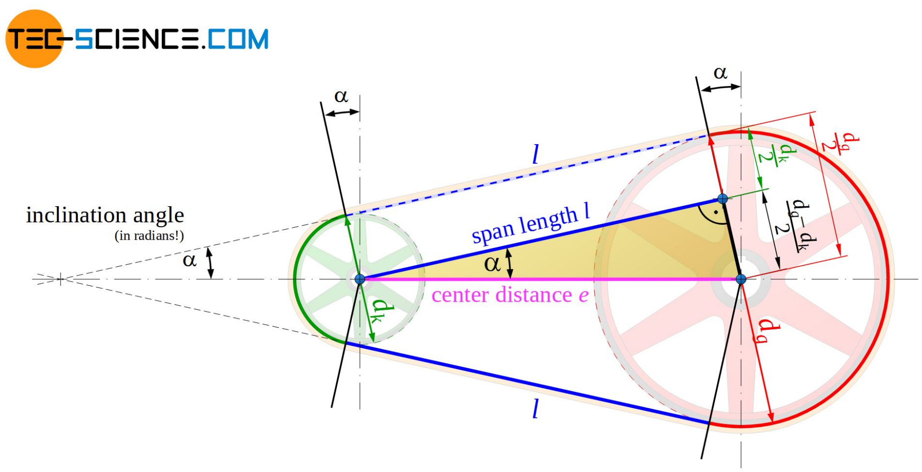

Figure 2: Belt geometry for two pulleys of radii r_p and r_s separated by center distance C. The belt wraps the pulleys with angles ϕ_p and ϕ_s and connects via two straight tangent spans.

Belt Length Decomposition The total belt length L_b is the sum of: (i) wrapped arc length on the primary pulley, (ii) wrapped arc length on the secondary pulley, and (iii) two straight tangent spans. Therefore,

$$L_b = r_p\phi_p + r_s\phi_s + 2\ell, \tag{8.3}$$

where ℓ is the length of a single straight tangent span segment.

Straight Span Length Let ℓ denote the length of the straight belt segment between tangent points. From the right-triangle geometry formed by the pulley centers and tangent points,

$$\ell = \sqrt{C^2 - (r_s - r_p)^2}. \tag{8.4}$$

Tangency Angle (Signed Convention) Let λ denote the signed angle between the line of centers and the straight tangent belt segment (as defined in Figure 2). We define

$$\sin\lambda = \frac{r_s - r_p}{C}, \tag{8.5}$$

and equivalently,

$$\lambda = \sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{8.6}$$

Under this convention, λ may be negative when $r_p > r_s$. This signed definition ensures the wrap distribution automatically adjusts when the pulley ordering reverses.

Wrap Angles The wrap angles on the primary and secondary pulleys are

$$\phi_p = \pi - 2\lambda, \tag{8.7}$$
$$\phi_s = \pi + 2\lambda. \tag{8.8}$$

If $\lambda < 0$, these expressions automatically redistribute wrap between pulleys while preserving total belt length.

Closed Form Belt-Length Expression Substituting Equation (8.4) and Equation (8.7)–Equation (8.8) into Equation (8.3) yields

$$\begin{aligned} L_b &= r_p(\pi - 2\lambda) + r_s(\pi + 2\lambda) + 2\sqrt{C^2 - (r_s - r_p)^2} \\ &= \pi(r_p + r_s) + 2\lambda(r_s - r_p) + 2\sqrt{C^2 - (r_s - r_p)^2}. \end{aligned} \tag{8.9}$$

Finally, substituting Equation (8.6) gives

$$L_b = \pi(r_p + r_s) + 2(r_s - r_p)\,\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2}. \tag{8.10}$$

Root Form (Geometric Compatibility Constraint) Rearranging Equation (8.10) yields

$$f(r_p, r_s, C) = \pi(r_p + r_s) + 2(r_s - r_p)\,\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2} - L_b. \tag{8.11}$$

The belt-length constraint is therefore

$$f(r_p, r_s, C) = 0. \tag{8.12}$$

Remark (Transcendental Nature and Approximation)

Equation Equation (8.12) is transcendental due to the inverse-sine and square-root terms and therefore does not admit a closed-form solution.
In this work, the constraint is solved numerically using a bracketed root-finding method (e.g., bisection or Brent's method), which is deterministic and robust over the physically admissible interval.
It is common in simplified treatments to assume

$$\frac{r_s - r_p}{C} \approx 0,$$

which implies $\lambda \approx 0$ and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this small-angle assumption, the belt-length equation reduces to a much simpler approximate form.
However, this approximation can introduce substantial geometric error when $\frac{r_s - r_p}{C}$ is not small. Appendix **??** quantifies this deviation and demonstrates that the approximation is not acceptable under realistic operating conditions.

8.5 Center-to-Center Distance Determination

The center-to-center distance C is a fixed geometric parameter of the assembled CVT system. Once the transmission is constructed, C does not vary during operation.

To determine C, we evaluate the belt-length constraint Equation (8.12) at the known initial configuration of the system.

Initial Geometric Configuration At startup, the CVT is assumed to be in its lowest ratio configuration:

- The primary pulley is at its minimum outer radius,

$$r_p = r_{p,\min}.$$

- The secondary pulley is at its maximum outer radius,

$$r_s = r_{s,\max}.$$

These values are determined by the physical geometry of the sheaves and represent the fully separated primary and fully closed secondary configuration.

Center Distance from Belt-Length Compatibility Substituting these known radii into Equation (8.12) yields

$$f(r_{p,\min}, r_{s,\max}, C) = 0. \tag{8.13}$$

This equation contains a single unknown, C, and is solved numerically to determine the fixed center-to-center distance consistent with the specified belt length L_b.

Remark (One-Time Geometric Solve)

The center distance C is determined once during model initialization. Thereafter, C is treated as a constant in all subsequent geometric and dynamic calculations.

8.6 Secondary Outer Radius as a Function of Shift

With the center-to-center distance C determined from Section 8.5, and the primary radius $r_p(s)$ defined by Equation (8.1), the secondary outer radius is obtained by enforcing the belt-length compatibility constraint.

Geometric Coupling At any shift position s, the primary radius $r_p(s)$ is known. The secondary radius r_s must therefore satisfy

$$f(r_p(s), r_s, C) = 0, \tag{8.14}$$

where $f(\cdot)$ is defined in Equation (8.11).

Numerical Determination of $r_s(s)$ Because Equation (8.14) is transcendental in r_s, the secondary radius is obtained numerically for each value of s:

$$r_{s,\text{outer}}(s) = \operatorname*{root}_{r_s} \; f(r_p(s), r_s, C)\,. \tag{8.15}$$

Remark (Dynamic Root Solve)

Unlike the center-distance calculation in Section 8.5, which is performed once during initialization, this root solve is performed repeatedly as the shift coordinate s evolves during simulation. Because the physically admissible interval for r_s is bounded, a bracketed solver (e.g., Brent's method) converges rapidly.

8.7 Effective Torque Radius

As noted previously, the belt possesses a finite radial thickness h_b. All geometric derivations were performed using the outer belt surface for convenience. However, torque is transmitted through the belt cross-section, and the effective moment arm corresponds approximately to the mid-height of the belt.

The effective torque radius for each pulley is therefore defined as

$$r_{p,\text{eff}} = r_{p,\text{outer}} - \frac{h_b}{2}, \tag{8.16}$$

$$r_{s,\text{eff}} = r_{s,\text{outer}} - \frac{h_b}{2}. \tag{8.17}$$

These effective radii are used in all subsequent torque transmission calculations.

8.8 CVT Ratio Definition

The instantaneous CVT ratio is defined as the ratio of secondary to primary effective radii:

$$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}. \tag{8.18}$$

Because $r_{p,\text{eff}}(s)$ increases with shift s, the ratio $R(s)$ decreases as the primary closes.

Thus:

- Small s (minimum primary radius) corresponds to high ratio.
- Large s (maximum primary radius) corresponds to low ratio.

The transmission ratio is therefore a purely geometric function of shift position.

8.9 Ratio Rate of Change

During shifting, the ratio R varies because the effective pulley radii vary with the shift coordinate s. This section derives an expression for $\dot{R}$ in terms of $\dot{s}$ and geometric derivatives.

8.9.1 Start from the Ratio Definition

The ratio is defined by the effective radii as

$$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}. \tag{8.19}$$

Because R depends on time only through $s(t)$, the chain rule gives

$$\dot{R} = \frac{dR}{ds}\dot{s}. \tag{8.20}$$

Thus, the objective is to compute dR/ds.

8.9.2 Differentiate R with Respect to s

Differentiating Equation (8.19) with respect to s (quotient rule) gives

$$\frac{dR}{ds} = \frac{d}{ds}\left(\frac{r_{s,\text{eff}}}{r_{p,\text{eff}}}\right) = \frac{\left(\frac{dr_{s,\text{eff}}}{ds}\right) r_{p,\text{eff}} - r_{s,\text{eff}}\left(\frac{dr_{p,\text{eff}}}{ds}\right)}{r_{p,\text{eff}}^2}. \tag{8.21}$$

Rearranging into a form that exposes the required components:

$$\frac{dR}{ds} = \frac{1}{r_{p,\text{eff}}}\frac{dr_{s,\text{eff}}}{ds} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2}\frac{dr_{p,\text{eff}}}{ds}. \tag{8.22}$$

At this point, the remaining task is to express the two derivatives $dr_{p,\text{eff}}/ds$ and $dr_{s,\text{eff}}/ds$ in computable form.

8.9.3 Eliminating dr_s/ds Using the Belt-Length Constraint

The secondary radius $r_s(s)$ is defined implicitly through the belt-length constraint and does not admit a closed-form expression in terms of r_p. Direct differentiation of $r_s(s)$ is therefore not practical.

Instead, we differentiate the belt-length compatibility constraint derived in Section 8.4, which is written as

$$f(r_p, r_s, C) = 0, \tag{8.23}$$

with $f(\cdot)$ defined explicitly in Equation (8.11) and C constant. Since $r_p = r_p(s)$ and $r_s = r_s(s)$, differentiating Equation (8.23) with respect to s gives (multivariable chain rule)

$$\frac{\partial f}{\partial r_p}\frac{dr_p}{ds} + \frac{\partial f}{\partial r_s}\frac{dr_s}{ds} = 0. \tag{8.24}$$

Solving Equation (8.24) for dr_s/ds yields

$$\frac{dr_s}{ds} = -\frac{\frac{\partial f}{\partial r_p}}{\frac{\partial f}{\partial r_s}}\frac{dr_p}{ds}. \tag{8.25}$$

8.9.4 Single Expression for dR/ds in Terms of Remaining Unknowns

Substituting Equation (8.25) into Equation (8.22) eliminates dr_s/ds and gives a single expression for dR/ds in terms of the belt constraint partial derivatives and the primary geometric derivative:

$$\frac{dR}{ds} = -\frac{1}{r_{p,\text{eff}}}\frac{\frac{\partial f}{\partial r_p}}{\frac{\partial f}{\partial r_s}}\frac{dr_p}{ds} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2}\frac{dr_{p,\text{eff}}}{ds}. \tag{8.26}$$

Equation Equation (8.26) makes the remaining requirements explicit. To evaluate dR/ds (and therefore $\dot{R}$ via Equation (8.20)), three quantities are required:

1. $\dfrac{\partial f}{\partial r_p}$,
2. $\dfrac{\partial f}{\partial r_s}$,
3. $\dfrac{dr_p}{ds}$ (and the associated $\dfrac{dr_{p,\text{eff}}}{ds}$).

The following subsections derive these three quantities for the specific belt-length function f defined in Equation (8.11) and for the primary geometry defined in Equation (8.1).

8.9.5 Partial Derivatives of the Belt Constraint

Recall the belt-length constraint written in root form as

$$f(r_p, r_s, C) = \pi(r_p + r_s) + 2(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2} - L_b = 0. \tag{8.27}$$

We now compute the two partial derivatives required in Equation (8.25), differentiating term-by-term and applying the product rule directly to the inverse-sine term.

Product rule for the inverse-sine factor Consider the factor

$$(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right).$$

Differentiating with respect to r_p (product rule) gives

$$\frac{\partial}{\partial r_p}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] = \underbrace{\frac{\partial}{\partial r_p}(r_s - r_p)}_{(1)}\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + (r_s - r_p)\underbrace{\frac{\partial}{\partial r_p}\sin^{-1}\left(\frac{r_s - r_p}{C}\right)}_{(2)}. \tag{8.28}$$

We therefore compute the two sub-derivatives.

Derivative (1)

$$\frac{\partial}{\partial r_p}(r_s - r_p) = -1. \tag{8.29}$$

Derivative (2) Using the chain rule,

$$\frac{d}{dx}\sin^{-1}(x) = \frac{1}{\sqrt{1 - x^2}},$$

let

$$u = \frac{r_s - r_p}{C}.$$

Then

$$\frac{\partial u}{\partial r_p} = -\frac{1}{C}.$$

Therefore,

$$\frac{\partial}{\partial r_p} \sin^{-1}\left(\frac{r_s - r_p}{C}\right) = \frac{1}{\sqrt{1-u^2}} \left(-\frac{1}{C}\right). \tag{8.30}$$

Now simplify the denominator:

$$\sqrt{1-u^2} = \sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2} = \frac{\sqrt{C^2 - (r_s - r_p)^2}}{C}.$$

Substituting into Equation (8.30) yields

$$\frac{\partial}{\partial r_p} \sin^{-1}\left(\frac{r_s - r_p}{C}\right) = -\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{8.31}$$

Substitute into the product rule Substituting Equation (8.29) and Equation (8.31) into Equation (8.28) gives

$$\begin{aligned}\frac{\partial}{\partial r_p} \left[(r_s - r_p) \sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] &= (-1) \sin^{-1}\left(\frac{r_s - r_p}{C}\right) + (r_s - r_p) \left(-\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}}\right) \\ &= -\sin^{-1}\left(\frac{r_s - r_p}{C}\right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}. \end{aligned} \tag{8.32}$$

Derivative of the square-root term For the straight-span contribution,

$$2\sqrt{C^2 - (r_s - r_p)^2},$$

we apply the chain rule:

$$\frac{\partial}{\partial r_p} \left[2\sqrt{C^2 - (r_s - r_p)^2}\right] = \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{8.33}$$

Assemble $\partial f / \partial r_p$ Differentiating Equation (8.27) term-by-term gives

$$\begin{aligned}\frac{\partial f}{\partial r_p} &= \pi + 2\left(-\sin^{-1}\left(\frac{r_s - r_p}{C}\right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}\right) + \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \end{aligned} \tag{8.34}$$

The square-root contributions cancel exactly.

Derivative with respect to r_s The same procedure yields

$$\frac{\partial}{\partial r_s}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] = \sin^{-1}\left(\frac{r_s - r_p}{C}\right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}, \tag{8.35}$$

and

$$\frac{\partial}{\partial r_s}\left[2\sqrt{C^2 - (r_s - r_p)^2}\right] = -\frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{8.36}$$

Assembling terms gives

$$\begin{aligned}\frac{\partial f}{\partial r_s} &= \pi + 2\left(\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}\right) - \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right).\end{aligned} \tag{8.37}$$

Summary of Belt Constraint Partial Derivatives From the derivations above, the partial derivatives of the belt-length constraint Equation (8.27) are

$$\frac{\partial f}{\partial r_p} = \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right), \tag{8.38}$$

$$\frac{\partial f}{\partial r_s} = \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{8.39}$$

These expressions are globally valid under the geometric definition of λ given in Equation (8.6) and require no piecewise case distinction.

8.9.6 Primary Radius Derivative

From the piecewise primary radius model Equation (8.1), the primary outer radius satisfies

$$r_{p,\text{outer}}(s) = \begin{cases} r_{p,\min}, & 0 \le s < s_{\text{dz}}, \\ r_{p,\min} + \dfrac{s - s_{\text{dz}}}{2\tan\beta}, & s_{\text{dz}} \le s \le s_{\max}. \end{cases}$$

Differentiating with respect to s gives

$$\frac{dr_p}{ds} = \frac{dr_{p,\text{outer}}}{ds} = \begin{cases} 0, & 0 \le s < s_{\text{dz}}, \\ \dfrac{1}{2\tan\beta}, & s_{\text{dz}} \le s \le s_{\max}. \end{cases} \tag{8.40}$$

Because $r_{p,\text{eff}} = r_{p,\text{outer}} - h_b/2$ and h_b is constant, the same derivative applies:

$$\frac{dr_{p,\text{eff}}}{ds} = \frac{dr_p}{ds}. \tag{8.41}$$

Remark (At the Travel Limit)

If the model clamps s at the physical limit $s_{\max}$, then $r_{p,\text{outer}}$ (and thus $r_{p,\text{eff}}$) is constant beyond this point and the derivative is zero for $s > s_{\max}$. In this document we restrict to the admissible domain $0 \le s \le s_{\max}$.

8.9.7 Final Expression for dR/ds and $\dot{R}$

Equation (8.26) expresses dR/ds in terms of three quantities: (i) the belt-constraint partial derivatives $\partial f/\partial r_p$ and $\partial f/\partial r_s$, and (ii) the primary geometric derivatives dr_p/ds and $dr_{p,\text{eff}}/ds$.

From Section 8.9.5, the partial derivatives of the belt-length constraint Equation (8.11) are

$$\frac{\partial f}{\partial r_p} = \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right), \tag{8.42}$$

$$\frac{\partial f}{\partial r_s} = \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{8.43}$$

From Section 8.9.6, the primary outer-radius derivative is Equation (8.40), and since $r_{p,\text{eff}} = r_{p,\text{outer}} - h_b/2$ with h_b constant, Equation (8.41) gives $dr_{p,\text{eff}}/ds = dr_p/ds$.

Substituting Equation (8.42)–Equation (8.43) and Equation (8.41) into Equation (8.26) yields

$$\frac{dR}{ds} = \left[-\frac{1}{r_{p,\text{eff}}}\frac{\pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right)} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2}\right]\frac{dr_p}{ds}. \tag{8.44}$$

Finally, substituting the piecewise primary derivative Equation (8.40) gives the fully expanded result

$$\frac{dR}{ds} = \begin{cases} 0, & 0 \le s < s_{\text{dz}}, \\ \dfrac{1}{2\tan\beta}\left[-\dfrac{1}{r_{p,\text{eff}}}\dfrac{\pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right)} - \dfrac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2}\right], & s_{\text{dz}} \le s \le s_{\max}. \end{cases} \tag{8.45}$$

Using the chain rule Equation (8.20), $\dot{R} = (dR/ds)\dot{s}$. Substituting Equation (8.45) and rearranging gives

$$\dot{R} = \begin{cases} 0, & 0 \le s < s_{\text{dz}}, \\ -\dfrac{\dot{s}}{2\tan\beta\, r_{p,\text{eff}}}\left[\dfrac{\pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right)} + \dfrac{r_{s,\text{eff}}}{r_{p,\text{eff}}}\right], & s_{\text{dz}} \le s \le s_{\max}. \end{cases} \tag{8.46}$$

Summary Equations Equation (8.45) and Equation (8.46) provide a complete closed-form expression for the geometric sensitivity of the transmission ratio and its time rate of change.

Specifically,

- $\dfrac{dR}{ds}$ quantifies how sensitive the CVT ratio is to axial sheave motion.
- $\dot{R} = \dfrac{dR}{dt}$ relates axial shift velocity $\dot{s}$ directly to rotational ratio dynamics.

The result captures three coupled geometric effects:

1. The primary sheave cone geometry through β,
2. The nonlinear belt-length compatibility through the inverse-sine term,
3. The redistribution of effective radii between pulleys during shift.

The deadzone behavior is explicitly represented: when $0 \le s < s_{\mathrm{dz}}$, the effective radii are constant and $\dot{R} = 0$.

In the active shift region $s_{\mathrm{dz}} \le s \le s_{\max}$, ratio dynamics are governed entirely by geometric coupling and axial shift velocity.

This completes the purely kinematic derivation of CVT ratio evolution. Subsequent sections incorporate force balance and torque transfer to determine $\dot{s}$ dynamically.

8.10 Section Summary and Forward Connection

This section has established a complete *geometric foundation* for the CVT model. The development was intentionally kinematic: no force balances, no slip assumptions, and no torque relations were introduced. The objective was to construct a mathematically closed mapping

$$s(t) \longrightarrow r_{p,\mathrm{eff}}(s),\ r_{s,\mathrm{eff}}(s) \longrightarrow R(s) \longrightarrow \dot{R}(s, \dot{s}),$$

with all intermediate relationships derived explicitly.

What Has Been Established The following key geometric components are now rigorously defined:

1. **Primary Radius Law** Equation (8.1) provides the piecewise relationship between axial shift s and primary outer radius, governed by the cone half-angle β and deadzone s_{dz}.
2. **Belt-Length Compatibility Constraint** Equation (8.12) defines the transcendental geometric coupling between r_p, r_s, and center distance C. This constraint enforces constant belt length and represents the core geometric coupling of the transmission.
3. **Center Distance Determination** Section 8.5 determines C once during initialization by solving Equation (8.13). After this step, C is fixed for the remainder of the model.
4. **Secondary Radius Implicit Law** Equation (8.14) and Equation (8.15) define $r_s(s)$ implicitly via the belt constraint, producing the geometric redistribution of radius during shift.
5. **Effective Torque Radii** Section 8.7 corrects outer radii to effective radii via Equation (8.2), ensuring torque calculations later act at the belt mid-height.

6. **Transmission Ratio** Equation (8.18) defines the purely geometric ratio

$$R(s) = \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}}.$$

7. **Ratio Sensitivity and Time Evolution** Section 8.9 culminates in the closed-form expressions Equation (8.45) and Equation (8.46), which provide

$$\frac{dR}{ds} \quad \text{and} \quad \dot{R} = \frac{dR}{ds}\dot{s}.$$

   These expressions fully capture the nonlinear geometric sensitivity introduced by belt-length compatibility and cone geometry.

Structural Importance for the Full Model This section establishes the geometric backbone of the entire CVT simulation framework. Every dynamic result that follows will rely on the mappings derived here.

In particular:

- Torque transfer equations will use $r_{p,\text{eff}}$ and $r_{s,\text{eff}}$ from Section 8.7.
- Axial force balances will determine $\dot{s}$ dynamically.
- The ratio dynamics in Equation (8.46) will convert axial motion into rotational acceleration coupling between shafts.
- Slip models and capstan friction relations will depend explicitly on wrap angles Equation (8.7)–Equation (8.8) and the geometric redistribution captured in Equation (8.11).
- Power flow and coupling torque formulations will use $R(s)$ and $\dot{R}$ directly to relate engine dynamics to driven-shaft dynamics.

Thus, this section isolates geometry from physics. All subsequent force and torque derivations will treat the geometric relationships derived here as fixed, differentiable mappings.

Key Modeling Properties Several important structural properties are now clear:

1. The ratio evolution is inherently nonlinear due to the inverse-sine term in the belt constraint.
2. The geometric coupling is globally valid (no small-angle approximation required).
3. The deadzone is explicitly represented: for $0 \leq s < s_{\text{dz}}$, R is constant.
4. Ratio sensitivity depends on both cone angle β and instantaneous pulley radii.
5. The system is fully differentiable within the admissible domain $s_{\text{dz}} \leq s \leq s_{\max}$, enabling continuous-time simulation and ODE integration.

Transition to Dynamics With geometry now fully specified, the model is ready to incorporate force balance and torque transfer.

The next major objective is to determine $\dot{s}$ from axial forces (centrifugal force, spring force, belt tension, and normal reaction forces). Once $\dot{s}$ is known, the ratio evolution Equation (8.46) immediately provides rotational coupling between shafts.

In other words:

$$\text{Axial Dynamics} \longrightarrow \dot{s} \longrightarrow \dot{R} \longrightarrow RotationalDynamics.$$

This completes the geometric phase of the CVT derivation. The model is now prepared for the introduction of force and torque equilibrium in the subsequent section.

9 Pulley Axial Force Models

This section introduces the axial force models that drive sheave motion and therefore determine the evolution of the shift coordinate $s(t)$.

Recall from the state definition section that the dynamic state vector contains

$$\left(\, s,\ \dot{s},\ \omega_p,\ \omega_s \,\right),$$

where s is the axial shift coordinate, $\dot{s}$ is the axial shift velocity, ω_p is the primary pulley angular velocity, and ω_s is the secondary pulley angular velocity. These quantities will be treated as the primary dynamic variables throughout the remainder of the model.

The goal of this section is to express the net axial forces generated by the primary and secondary pulley mechanisms as functions of the current state. These forces will later be combined using Theory 1 (Newton's Second Law (Translation)) to obtain an equation of motion for s (and therefore $\dot{s}$ and $\ddot{s}$).

Although the specific models developed here correspond to the hardware considered in this work (flyweight/ramp actuation on the primary and a torque-reactive helix with springs on the secondary), the formulation is modular: alternative actuator models may be substituted provided they return the correct net axial force.

9.1 Ramp Geometry Parameterization

Both pulley actuators redirect forces through inclined surfaces. To keep the force models general and drop-in replaceable, we represent each mechanism's geometry using user-supplied functions of the axial shift coordinate s.

In this document, s is the *axial* displacement along the shaft axis (see Section 8). A larger s corresponds to a more-closed configuration of the active sheave.

Primary Ramp Geometry (Axial-to-Radial Profile) On the primary pulley, centrifugal (radial) force is redirected into axial closing force through a ramp. We parameterize the ramp by an axial-to-radial profile

$$r_f = r_f(s), \tag{9.1}$$

where $r_f(s)$ is the effective flyweight radius.

Because s is axial distance, the local ramp slope is dr_f/ds. The corresponding ramp angle is defined consistently as

$$\alpha_p(s) \;\equiv\; \tan^{-1}\!\left(\frac{dr_f}{ds}\right), \qquad \text{so that} \qquad \tan\alpha_p(s) = \frac{dr_f}{ds}. \tag{9.2}$$

This definition is physically intuitive: a steeper ramp yields a larger change in radius per unit axial travel and therefore larger axial gain through the $\tan \alpha_p$ factor used later in the force model.

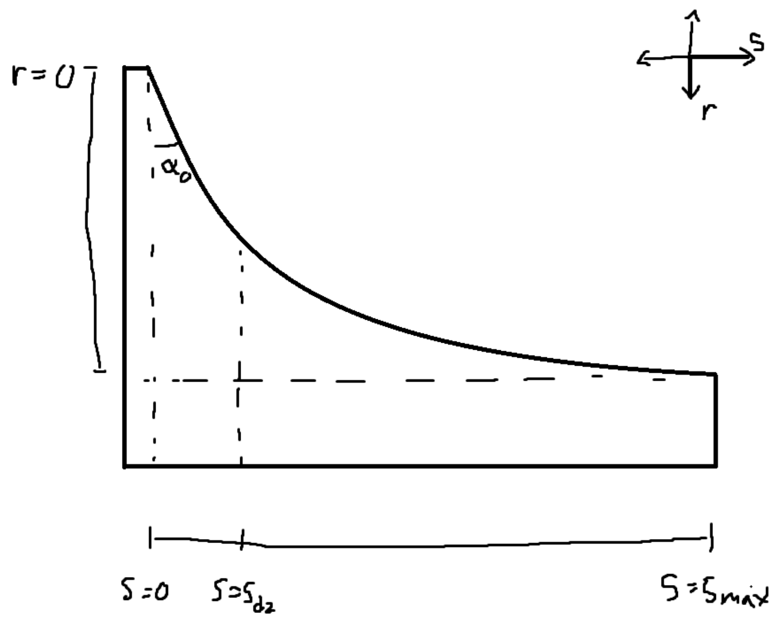

Figure 3: Primary ramp geometry. The flyweight radius is parameterized by $r_f(s)$, where s is axial travel. The local ramp angle is defined by $\tan \alpha_p(s) = dr_f/ds$, so steeper profiles (larger dr_f/ds) redirect a larger fraction of radial force into axial closing force.

Primary Admissibility Conditions The primary ramp profile $r_f(s)$ must satisfy:

1. $r_f(s) \in C^1([0, s_{\max}])$ (continuously differentiable),
2. $\dfrac{dr_f}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\dfrac{dr_f}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the ramp angle $\alpha_p(s)$ is well-defined and continuous. Condition (2) guarantees that increasing axial displacement does not reduce flyweight radius. Condition (3) ensures the ramp angle remains strictly below $\pi/2$ and the axial gain $\tan \alpha_p$ remains finite.

Secondary Helix Geometry (Axial-to-Rotation Mapping) On the secondary pulley, axial travel produces a relative rotation of the helix mechanism, loading the torsional spring and generating torque-reactive clamping. We parameterize this kinematically as

$$\theta_s = \theta_s(s), \tag{9.3}$$

where $\theta_s(s)$ is the helix-induced rotation.

The quantity $d\theta_s/ds$ is the helix "lead" (rotation per unit axial travel). Under the convention used in this work, *shallower* helix ramps produce *more* rotation for the same axial travel, while *steeper* ramps produce *less* rotation. To encode this consistently, we define the effective helix angle $\alpha_s(s)$ by the inverse-slope relationship

$$\tan \alpha_s(s) \;\equiv\; \frac{1}{r_h \, \dfrac{d\theta_s}{ds}}, \tag{9.4}$$

where r_h is the effective helix radius. With this definition, a decrease in α_s (a "shallower" ramp) corresponds to an increase in $d\theta_s/ds$, matching the physical behavior described above.

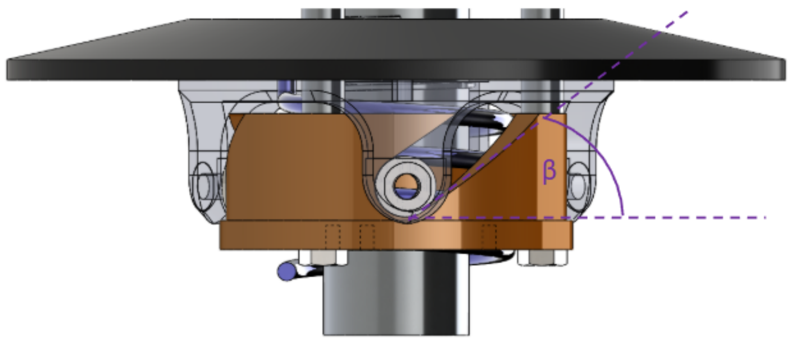

Figure 4: Secondary helix geometry. Axial travel s induces rotation $\theta_s(s)$ of the helix mechanism. The helix angle is defined so that shallower ramps produce greater rotation per unit axial travel: $\tan \alpha_s(s) = \big(r_h \, d\theta_s/ds\big)^{-1}$.

Secondary Admissibility Conditions The helix rotation mapping $\theta_s(s)$ must satisfy:

1. $\theta_s(s) \in C^1([0, s_{\max}])$,
2. $\dfrac{d\theta_s}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\dfrac{d\theta_s}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the torque-to-axial conversion remains continuous. Condition (2) guarantees that increasing axial displacement increases torsional loading. Condition (3) ensures the effective helix angle remains strictly below $\pi/2$ and the geometric gain remains finite.

Remark (Hardware Abstraction)

The functions $r_f(s)$ and $\theta_s(s)$ encode the mechanism-specific ramp and helix geometry. The remaining axial-force equations depend on these mappings only through their local slopes (dr_f/ds and $d\theta_s/ds$), which provides a tight and self-consistent parameterization.
Derivations for common ramp and helix profiles are provided in Appendix **??**.

9.2 Primary Axial Force Model

The primary axial clamping force arises from centrifugal loading of the flyweights and opposition from the primary compression spring.

The state variable governing primary rotation is ω_p.

Centrifugal Loading of Flyweights From Theory 3 (Centrifugal Force),

$$F_c = m\omega^2 r.$$

Let

- m_f denote the flyweight mass,
- $r_f(s)$ denote the effective flyweight radius.

The radial centrifugal force is therefore

$$F_{p,\text{rad}}(s, \omega_p) = m_f \omega_p^2 r_f(s). \tag{9.5}$$

Radial-to-Axial Projection The ramp redirects radial force into axial force. From elementary trigonometry (see Figure 5), the axial component of a force acting along the ramp is obtained by

$$F_{p,\text{axial component}} = F_{p,\text{rad}} \tan \alpha_p(s). \tag{9.6}$$

From Section 9.1, the ramp angle satisfies

$$\tan \alpha_p(s) = \frac{dr_f}{ds}.$$

Substituting this into Equation (9.6) gives

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 r_f(s) \frac{dr_f}{ds}. \tag{9.7}$$

Thus the axial gain of centrifugal force is directly governed by the local slope of the ramp profile.

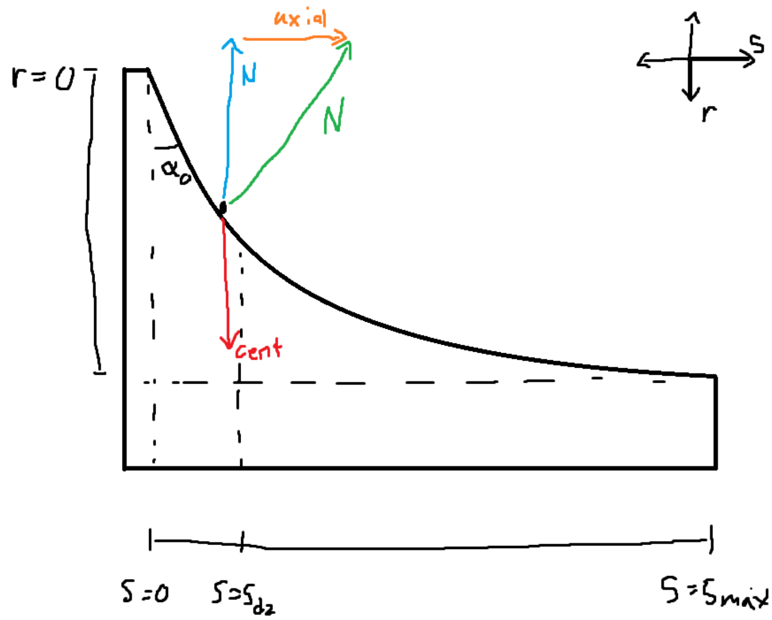

Figure 5: Primary ramp geometry. Radial centrifugal force is redirected into axial force through the ramp. The projection introduces a factor of $\tan \alpha_p$, which equals dr_f/ds by definition of the ramp slope.

Primary Compression Spring From Theory 4 (Hooke's Law (Compressional)),

$$F = k\Delta x.$$

Let k_p denote primary stiffness and $x_{p,0}$ preload.

$$F_{p,\text{spring}}(s) = k_p(x_{p,0} + s). \tag{9.8}$$

Net Primary Axial Force

$$F_{p,\text{ax}}(s, \omega_p) = m_f \omega_p^2 r_f(s) \frac{dr_f}{ds} - k_p(x_{p,0} + s). \tag{9.9}$$

9.3 Secondary Axial Force Model

The secondary pulley is torque-reactive. Axial clamping force is generated by torque about the shaft, which is redirected through the helix geometry, and is supplemented by a secondary compression spring.

Unlike the primary, the secondary axial force depends directly on shaft torque and therefore couples axial and rotational dynamics.

Definition of Secondary Shaft Torque Let τ_s denote the torque reacted at the secondary shaft. It will later be determined from the drivetrain torque balance using Theory 2 (Newton's Second Law (Rotation)).

Torsional Spring Torque From Theory 5 (Hooke's Law (Torsional)),

$$\tau = k\Delta\theta.$$

Let $k_{s,\theta}$ denote torsional stiffness and $\theta_{s,0}$ preload.

$$\tau_{s,\text{spring}}(s) = k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big). \tag{9.10}$$

Total Torque Applied to Helix

$$\tau_{s,\text{total}} = \tau_s + \tau_{s,\text{spring}}(s). \tag{9.11}$$

Torque-to-Axial Conversion Through the Helix From torque definition, $\tau = F_t r_h$, so

$$F_t = \frac{\tau_{s,\text{total}}}{r_h}.$$

The helix redirects tangential force into axial force. From trigonometry (see Figure 6),

$$F_{s,\text{helix}} = \frac{F_t}{2\tan\alpha_s(s)}. \tag{9.12}$$

From Section 9.1, the helix angle satisfies

$$\tan\alpha_s(s) = \frac{1}{r_h \dfrac{d\theta_s}{ds}}, \qquad \Rightarrow \qquad \frac{1}{\tan\alpha_s(s)} = r_h \frac{d\theta_s}{ds}.$$

Substituting into Equation (9.12) yields

$$F_{s,\text{helix}}(s, \tau_s) = \frac{\tau_{s,\text{total}}}{2} \frac{d\theta_s}{ds}. \tag{9.13}$$

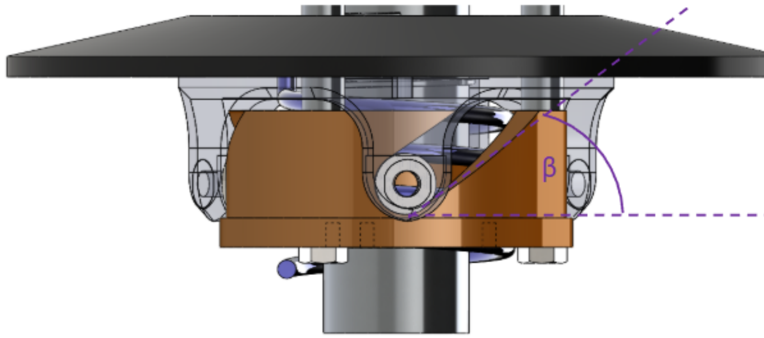

Figure 6: Secondary helix geometry. Tangential force generated by shaft torque is redirected into axial force. The projection introduces a factor $1/\tan\alpha_s$, which equals $r_h\, d\theta_s/ds$ by definition of the helix mapping.

Secondary Compression Spring In addition to helix loading, the secondary includes a compression spring. From Theory 4 (Hooke's Law (Compressional)),

$$F = k\Delta x.$$

Let $k_{s,x}$ denote the secondary compression stiffness and $x_{s,0}$ preload. With compression increasing with s,

$$F_{s,\mathrm{comp}}(s) = k_{s,x}(x_{s,0} + s). \tag{9.14}$$

Net Secondary Axial Force The net secondary axial force is the sum of the helix-generated clamping force and the compression spring force:

$$F_{s,\mathrm{ax}}(s, \tau_s) = \underbrace{\frac{\tau_s + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)}{2}\frac{d\theta_s}{ds}}_{\text{helix + torsional spring}} + \underbrace{k_{s,x}(x_{s,0} + s)}_{\text{compression spring}}. \tag{9.15}$$

9.4 Belt Centrifugal Force on the Wrap

In addition to actuator and spring forces, the belt itself generates a distributed centrifugal load while rotating around each pulley. This load acts radially outward along the wrapped belt segment and increases the normal loading between belt and sheaves. Following the axial–radial conversion discussed previously (see Figure 1), this contribution can be mapped into an effective axial opening force on each pulley.

Distributed Load Model Let the belt material have density ρ_b and cross-sectional area A_b (see Figure 8), giving a linear mass density

$$\mu_b = \rho_b A_b. \tag{9.16}$$

Consider a belt element on a pulley at angular position θ (measured about the pulley center), where the belt follows a circular arc of radius r over a wrap angle ϕ. The element has differential length

$$\mathrm{d}\ell = r\ \mathrm{d}\theta,$$

and differential mass

$$\mathrm{d}m = \mu_b\ \mathrm{d}\ell = \rho_b A_b\, r\ \mathrm{d}\theta. \tag{9.17}$$

From Theory 3 (Centrifugal Force), the centrifugal force on this element is

$$\mathrm{d}F_c = \ \mathrm{d}m\, \omega^2 r = \rho_b A_b\, r\ \mathrm{d}\theta\ \omega^2 r = \rho_b A_b\, \omega^2\, r^2\ \mathrm{d}\theta. \tag{9.18}$$

This force is directed radially outward and is distributed continuously around the wrap, as illustrated in Figure 7.

Figure 7: Distributed belt centrifugal force acting radially outward along the wrapped belt segment. The integral is taken over $\theta \in [-\phi/2,\ \phi/2]$.

Figure 8: Belt cross-section used to define area A_b (and therefore linear density $\mu_b = \rho_b A_b$).

Resultant Centrifugal Load Over a Wrap Angle Integrating Equation (9.18) over the wrap, taking the symmetric bounds $\theta \in [-\phi/2,\ \phi/2]$, yields the net radial centrifugal load on that pulley:

$$\begin{aligned} F_{c,\mathrm{rad}} &= \int_{-\phi/2}^{\phi/2} \rho_b A_b\, \omega^2\, r^2\ \mathrm{d}\theta = \rho_b A_b\, \omega^2\, r^2 \int_{-\phi/2}^{\phi/2} \mathrm{d}\theta \\ &= \rho_b A_b\, \omega^2\, r^2\, \phi. \end{aligned} \tag{9.19}$$

This expression assumes the belt rides at approximately constant radius r along the wrapped arc, which is consistent with the pulley effective-radius representation used in Section 8.

Axial Conversion Through the Sheave Cone The distributed centrifugal load acts radially and therefore must be converted into an equivalent axial closing force on the sheaves. Using the same axial–radial coupling as in the pulley geometry, the conjugate force mapping is

$$F_{c,\mathrm{ax}} = F_{c,\mathrm{rad}}\, \frac{\mathrm{d}r}{\mathrm{d}s}. \tag{9.20}$$

For a conical sheave with half-angle β (see Section 8.3), the kinematic relation is

$$\frac{\mathrm{d}r}{\mathrm{d}s} = \frac{1}{2\tan\beta},$$

so

$$F_{c,\mathrm{ax}} = \frac{F_{c,\mathrm{rad}}}{2\tan\beta} = \frac{\rho_b A_b\, \omega^2\, r^2\, \phi}{2\tan\beta}. \tag{9.21}$$

Application to Primary and Secondary Pulleys Applying Equation (9.21) to each pulley gives the centrifugal belt contribution to axial clamping:

$$F_{c,p}(s,\omega_p) = \frac{\rho_b A_b\, \omega_p^2\, r_{p,\text{out}}(s)^2\, \phi_p(s)}{2\tan\beta_p}, \tag{9.22}$$

$$F_{c,s}(s,\omega_s) = \frac{\rho_b A_b\, \omega_s^2\, r_{s,\text{out}}(s)^2\, \phi_s(s)}{2\tan\beta_s}, \tag{9.23}$$

where $\phi_p(s)$ and $\phi_s(s)$ are the wrap angles from Section 8.4 (e.g., Equation (8.7)–Equation (8.8)), and $r_{p,\text{out}}(s)$, $r_{s,\text{out}}(s)$ are the outer belt radii from Section 8.3 and Section 8.6. If desired, r_{out} may be replaced by the effective radius r_{eff} depending on where you want the belt mass centroid to act (addressed explicitly later in the belt mass distribution assumptions).

Remark (How This Will Be Used Later)

The terms $F_{c,p}$ and $F_{c,s}$ act as additional closing contributions and will be added to the actuator-generated axial forces in the net axial force balance. In the next section, these contributions will be combined and substituted into Theory 1 (Newton's Second Law (Translation)) to obtain the equation of motion for $s(t)$.

9.5 Axial Shift Dynamics

Axial Shift Equation of Motion

$$m_{\text{shift}}\ddot{s} = F_{p,\text{ax}} - F_{s,\text{ax}}$$

10 Belt Forces and Torque Transmission

10.1 Axial-to-Radial Force Conversion

$$F_{r,p} = \cdots \qquad F_{r,s} = \cdots \tag{10.1}$$

10.2 Belt Tension Distribution

Using **??** 9,

$$\frac{T_1}{T_2} = e^{\mu\beta} \tag{10.2}$$

10.3 Maximum Transferable Torque

Maximum Transferable Torque

$$T_{\max,p} = \cdots \qquad T_{\max,s} = \cdots$$

11 Dynamics: No-Slip Baseline

11.1 Coupling Torque

Coupling Torque Definition

$$T_c = \cdots$$

11.2 Vehicle Longitudinal Dynamics

$$m\dot{v} = \frac{T_{\text{wheel}}}{r_{\text{wheel}}} - F_{\text{resist}} \tag{11.1}$$

11.3 Final Coupled State-Space System

State-Space Form

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, u)$$

12 Slip Model

12.1 Slip Condition and Torque Saturation

$$|T_c| \leq T_{\max} \Rightarrow \text{no slip} \tag{12.1}$$

$$T = T_{\max}\ \text{sgn}(T_c) \Rightarrow \text{slip} \tag{12.2}$$

12.2 Slip Velocity

$$v_{\text{slip}} = r_p \omega_p - r_s \omega_s \tag{12.3}$$

A Parameter Tables

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